Lab 4: Estimates and confidence intervals

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This lab uses the `fitdistrplus` package for fitting distributions.

Reference: fitdistrplus: An R Package for Fitting Distributions

First, we need to install the package

```
install.packages("fitdistrplus")
```

if you don't have it.

Moreover we need to load the package because it is not automatically loaded.

To do this, use the `library()` function

```
library(fitdistrplus)
```

To get a list of the contents of the package, as well as help with respect to the package itself, use the help function

```
help(package="fitdistrplus")
```

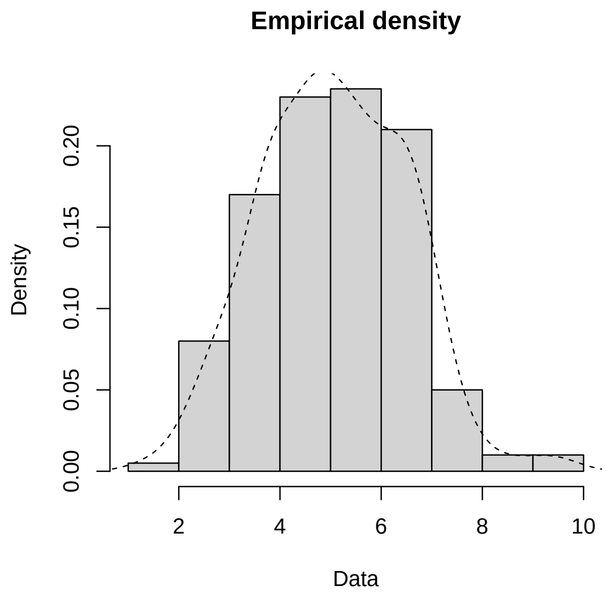
Simulated example

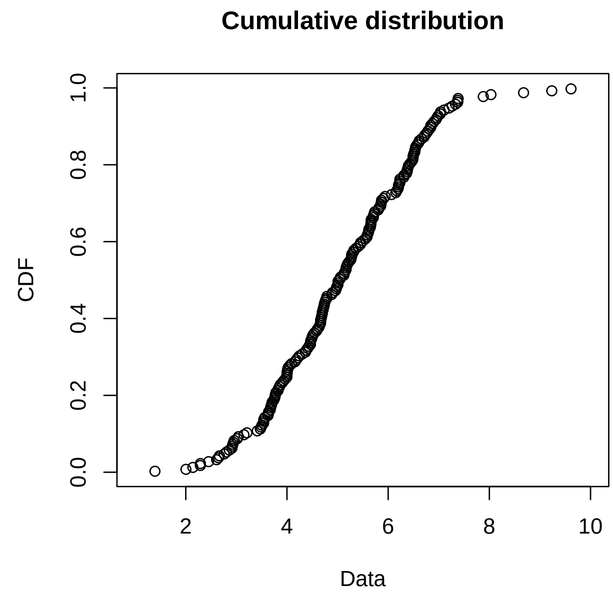
We consider 200 independent values from a random variable with Gaussian distribution with expectation 5 and variance 2, $\mathcal{N}(5, 2)$

```
n<-200
y<-rnorm(n,mean = 5, sd = sqrt(2))
```

We can first plot the empirical density and the histogram to gain insight of the data:

```
plotdist(y, histo = TRUE, demp = TRUE)
```





Fitting a distribution

We choose the Gaussian and Gamma distribution as statistical model:

```
fit.n  <- fitdist(y, distr = "norm")
fit.g  <- fitdist(y, distr = "gamma")
```

The default method is the MLE method. We can modify this choice method and switch to moment method using

```
fit.gm  <- fitdist(y, distr = "gamma",method = "mme")
```

We inspect the estimation results

```
summary(fit.g)
```

```
## Fitting of the distribution ' gamma ' by maximum likelihood
## Parameters :
##        estimate Std. Error
## shape 11.520289   1.135736
## rate   2.269963   0.228728
## Loglikelihood:  -358.2997   AIC:  720.5994   BIC:  727.196
## Correlation matrix:
##           shape      rate
## shape 1.0000000 0.9783931
## rate  0.9783931 1.0000000
```

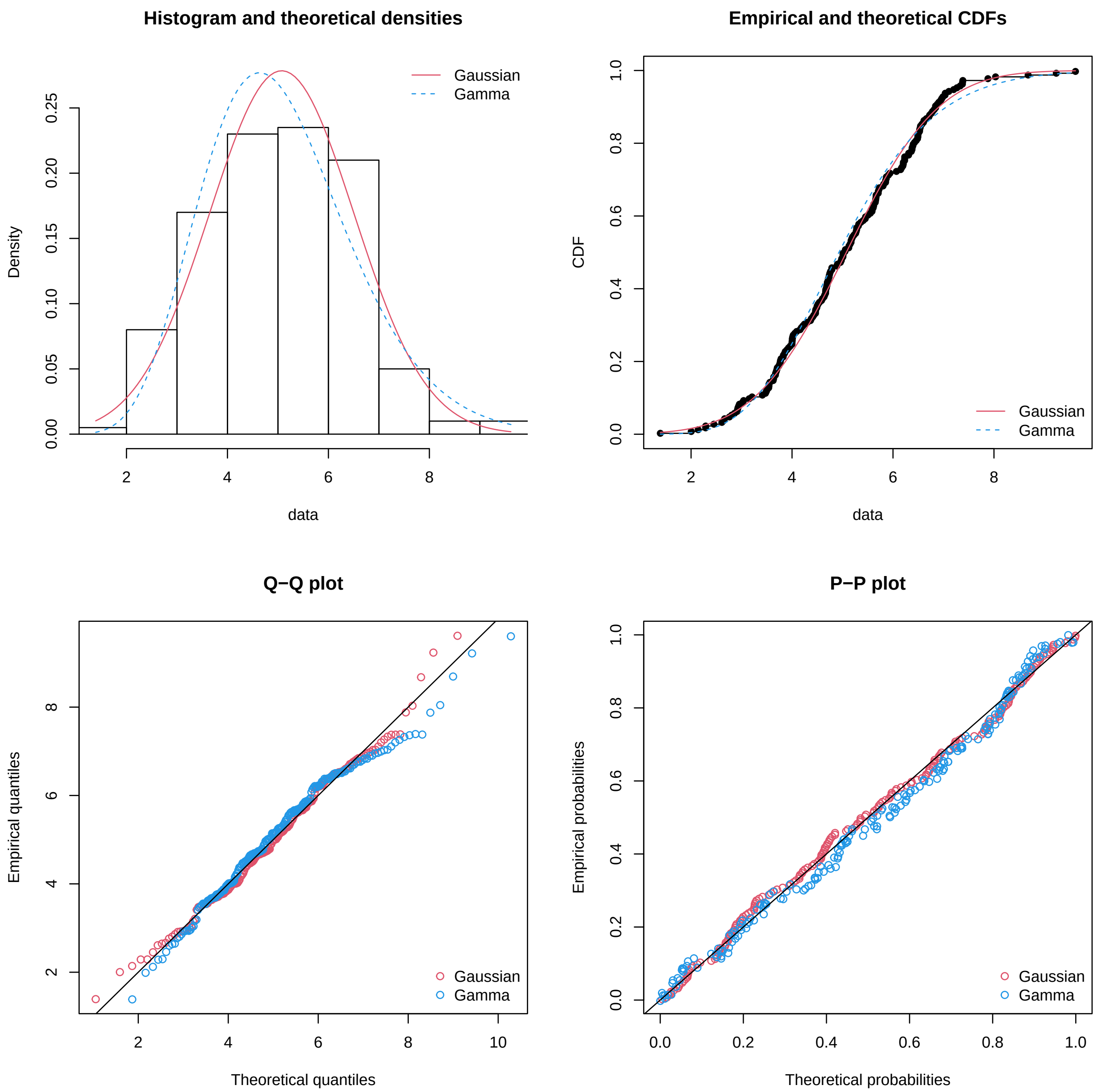
```
summary(fit.gm)
```

```
## Fitting of the distribution ' gamma ' by matching moments
## Parameters :
##        estimate Std. Error
## shape 12.552884  1.3043312
## rate   2.473397  0.2617009
## Loglikelihood:  -358.6861   AIC:  721.3723   BIC:  727.9689
## Correlation matrix:
##          shape     rate
## shape 1.000000 0.982049
```

```
## rate  0.982049 1.000000
```

We can assess graphically the goodness-of-fit

```r
par(mfrow=c(2,2))
plot.legend <- c("Gaussian", "Gamma")
estimated.model<-list(fit.n,fit.g)
denscomp(estimated.model, legendtext = plot.legend,fitcol=c(2,4))
cdfcomp (estimated.model, legendtext = plot.legend,fitcol=c(2,4))
qqcomp  (estimated.model, legendtext = plot.legend,fitcol=c(2,4))
ppcomp  (estimated.model, legendtext = plot.legend,fitcol=c(2,4))
```




```r
par(mfrow=c(1,1))
```

The package also provides some goodness-of-it statistics:

```r
gofstat(estimated.model, fitnames = plot.legend)
```

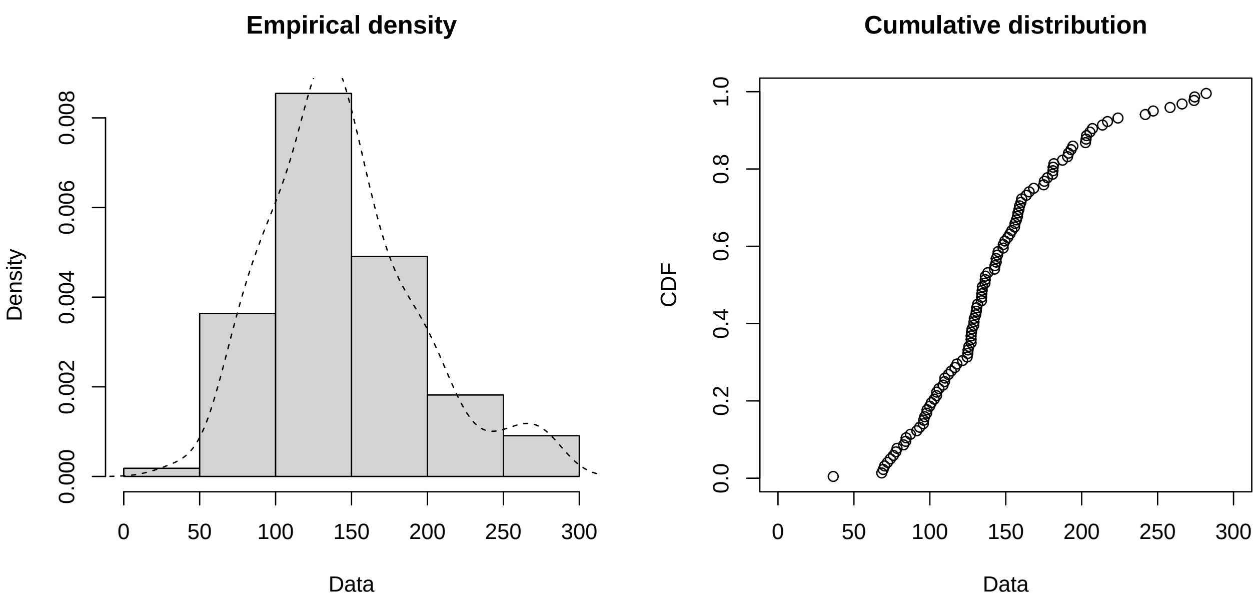
```
## Goodness-of-fit statistics
##                               Gaussian      Gamma
## Kolmogorov-Smirnov statistic 0.04864875 0.05528277
## Cramer-von Mises statistic   0.06592267 0.14057586
## Anderson-Darling statistic   0.46339935 0.96296785
##
## Goodness-of-fit criteria
##                                Gaussian    Gamma
## Akaike's Information Criterion 715.3284 720.5994
## Bayesian Information Criterion 721.9251 727.1960
```

Real data example

Let's use the winter precipitation at Basel station in Switzerland. Values are seasonal precipitation totals and units are in millimeters.

```r
basel<-read.table(file="basel-rainfall.txt",header=TRUE)
y<-basel$mm
```

```r
plotdist(y, histo = TRUE, demp = TRUE)
```



Kurtosis

Kurtosis is a statistical measure that describes the shape of a distribution's tails, indicating how frequently extreme values (outliers) occur.

The kurtosis index K is calculated as:

$$K = \frac{E[(Y-\mu)^4]}{\sigma^4}$$

Where:

- μ = mean of the distribution

- σ = standard deviation

For a sample of n values, a method of moments estimator of the population kurtosis can be defined as

$$\hat{K} = \frac{\frac{1}{n}\sum_{i=1}^{n}(y_i - \overline{y})^4}{\left[\frac{1}{n}\sum_{i=1}^{n}(y_i - \overline{y})^2\right]^2}$$

Types Based on Kurtosis Index

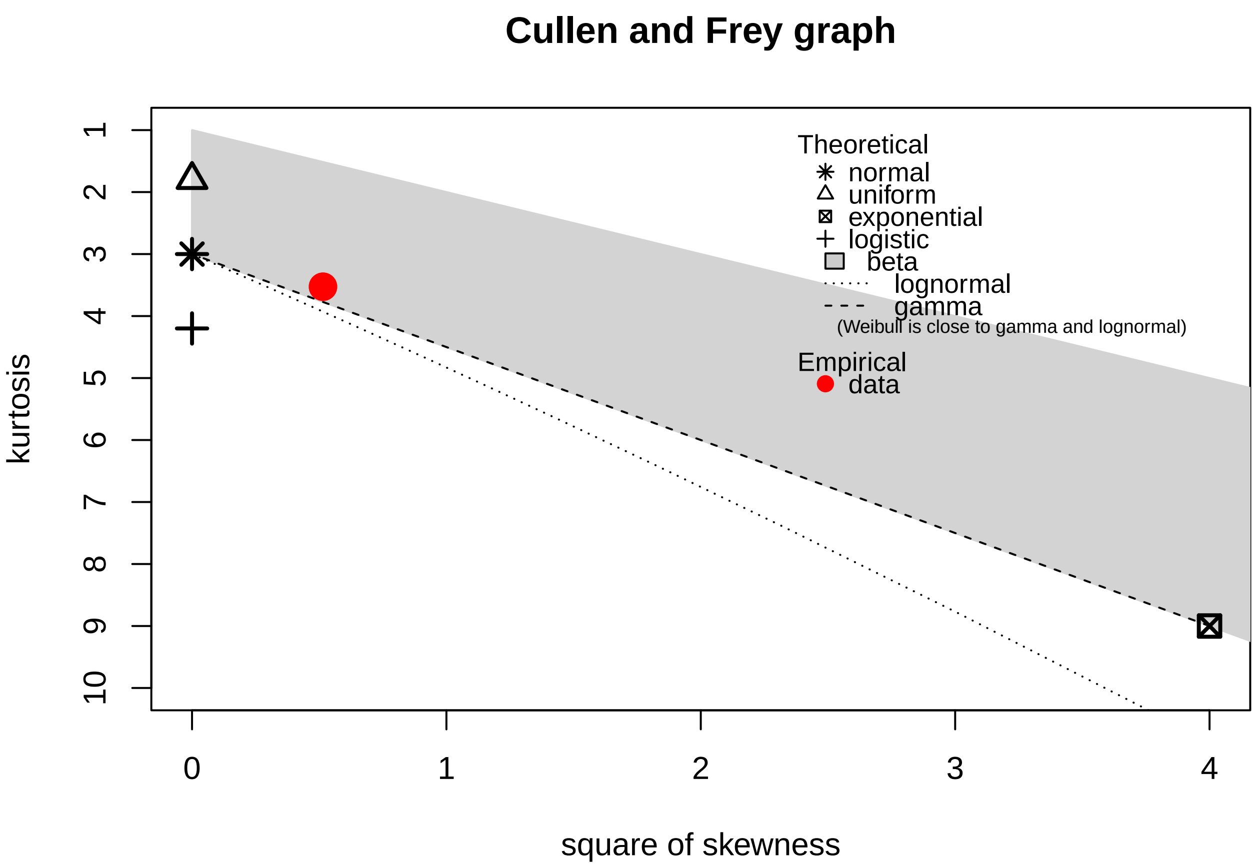
Type	Kurtosis Index	Excess Kurtosis	Description
Mesokurtic	$\simeq 3$	0	Normal distribution
Leptokurtic	> 3	> 0	Heavy tails, more extreme outliers
Platykurtic	< 3	< 0	Light tails, fewer extreme outliers

Interpretation

- **Higher kurtosis index** → more data in the tails (more outliers)
- **Lower kurtosis index** → less data in the tails (fewer outliers)
- **Excess kurtosis** is often used to compare against the normal distribution baseline of 0

The `descdist` function provides classical descriptive statistics (minimum, maximum, median, mean, standard deviation), skewness and kurtosis. A skewness-kurtosis plot such as the one proposed by Cullen and Frey (1999) is provided by the descdist function for the empirical distribution. On this plot, values for common distributions are displayed in order to help the choice of distributions to fit to data. For some distributions (normal, uniform, logistic, exponential), there is only one possible value for the skewness and the kurtosis. Thus, the distribution is represented by a single point on the plot. For other distributions, areas of possible values are represented, consisting in lines (as for gamma and lognormal distributions), or larger areas (as for beta distribution).

```r
descdist(y,discrete = FALSE)
```




```
## summary statistics
## ------
## min:  36.4   max:  282
## median:  135.45
## mean:  144.0082
## estimated sd:  49.20113
## estimated skewness:  0.7174984
## estimated kurtosis:  3.526643
```

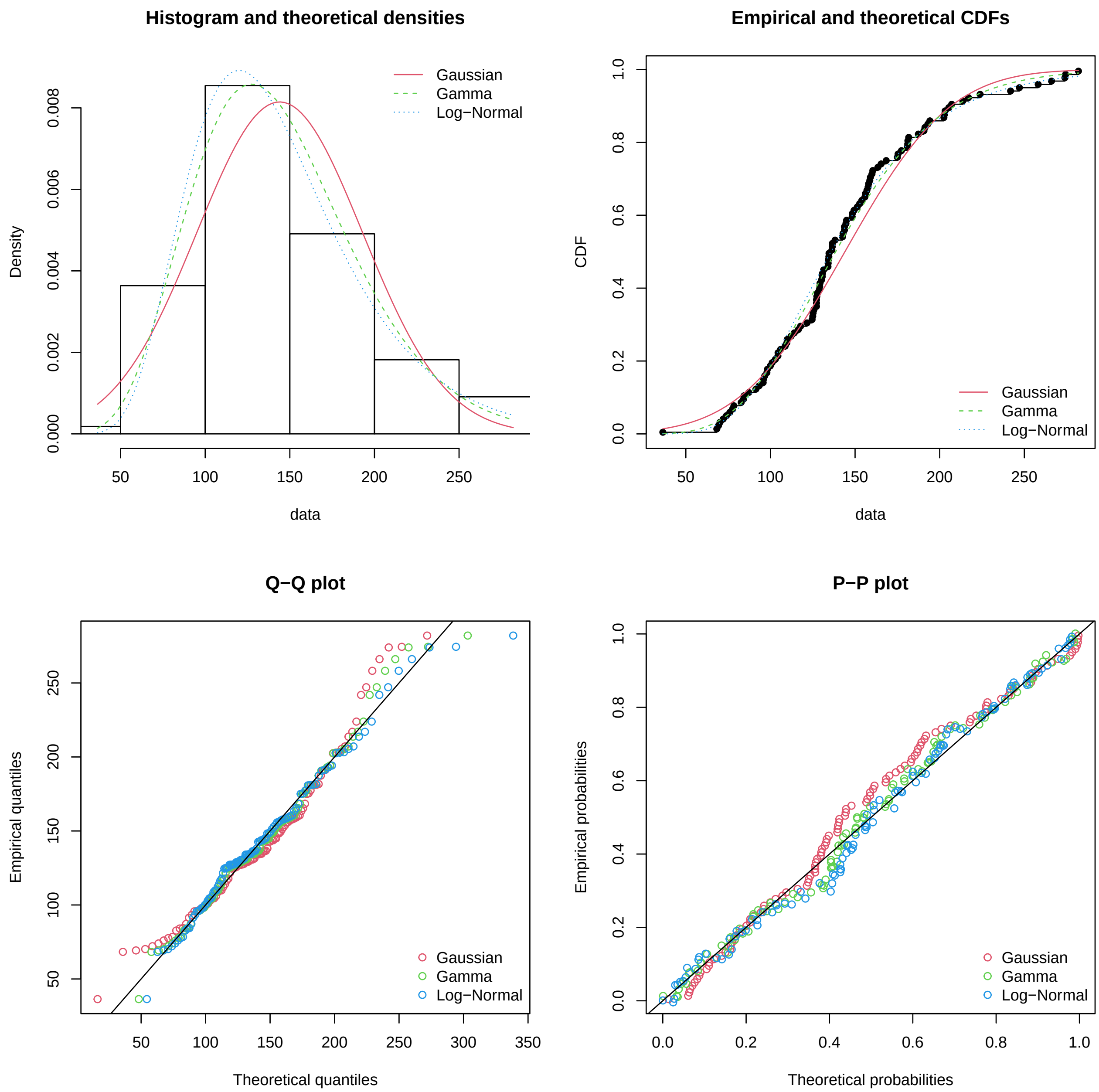
Fitting a distribution

We choose the Gaussian, the Gamma and the Log-Normal distribution as statistical model:

```
fit.n  <- fitdist(y, distr = "norm")
fit.g  <- fitdist(y, distr = "gamma")
fit.ln <- fitdist(y, distr = "lnorm")
```

We can assess graphically the goodness-of-fit

```
par(mfrow=c(2,2))
plot.legend <- c("Gaussian", "Gamma","Log-Normal")
estimated.model<-list(fit.n, fit.g,fit.ln)
denscomp(estimated.model, legendtext = plot.legend,fitcol=c(2,3,4))
cdfcomp (estimated.model, legendtext = plot.legend,fitcol=c(2,3,4))
qqcomp  (estimated.model, legendtext = plot.legend,fitcol=c(2,3,4))
ppcomp  (estimated.model, legendtext = plot.legend,fitcol=c(2,3,4))
```




```r
par(mfrow=c(1,1))
```

```r
gofstat(estimated.model, fitnames = plot.legend)
```

```
## Goodness-of-fit statistics
##                                Gaussian      Gamma Log-Normal
## Kolmogorov-Smirnov statistic 0.09543448 0.07231980 0.09387145
## Cramer-von Mises statistic   0.18909204 0.05918861 0.07689278
## Anderson-Darling statistic   1.19416447 0.35008578 0.42144857
##
## Goodness-of-fit criteria
##                                Gaussian    Gamma Log-Normal
## Akaike's Information Criterion 1172.264 1163.170   1165.756
## Bayesian Information Criterion 1177.664 1168.571   1171.157
```

Visually, and using the previous statistics, it seems that the gamma distribution is the preferred one among the candidates we chose to explore.

It could be useful to estimate a quantile of the distribution. This can be easily done using the quantile generic function defined for an object of class 'fitdist'.

```
quantile(fit.g,probs=0.4)
```

```
## Estimated quantiles for each specified probability (non-censored data)
##             p=0.4
## estimate 126.7679
```

which is equivalent to

```
qgamma(0.4,shape =fit.g$estimate[1],rate =fit.g$estimate[2])
```

```
## [1] 126.7679
```

We can compare the fitted quantile with the empirical one

```
quantile(y,probs=0.4)
```

```
##    40%
## 129.02
```

Confidence interval

```
alpha<-0.05
q<-qnorm(1-alpha/2)
std.alpha<-fit.g$sd[1]
alpha.hat<-fit.g$estimate[1]
c(alpha.hat-q*std.alpha,alpha.hat+q*std.alpha)
```

```
##     shape     shape
##  6.433727 10.921117
```

```
std.rate<-fit.g$sd[2]
rate.hat<-fit.g$estimate[2]
c(rate.hat-q*std.rate,rate.hat+q*std.rate)
```

```
##       rate       rate
## 0.04421880 0.07628745
```

Bootstraping the estimates (optional)

We can apply a non parametric bootstrap to estimate the uncertainty in the parameters. Argument `niter` sets the number of bootstrap replications.

```
ests.np <- bootdist(fit.g, niter = 1000,bootmethod = "nonparam")
summary(ests.np)
```

```
## Nonparametric bootstrap medians and 95% percentile CI
##           Median       2.5%       97.5%
## shape 8.90041098 6.87332372 11.92705756
## rate  0.06184018 0.04668281  0.08327432
```

Parametric bootstrap.

```
ests.p <- bootdist(fit.g, niter = 1000,bootmethod = "param")
summary(ests.p)
```

```
## Parametric bootstrap medians and 95% percentile CI
##           Median       2.5%       97.5%
## shape 8.70893374 6.76858934 11.49906421
## rate  0.06051935 0.04648517  0.08053194
```

95% percentile bootstrap confidence interval for a specific quantile, in this example $q_{0.99}$

```
qgamma(0.99,shape = fit.g$estimate[1],rate = fit.g$estimate[2])
```

```
## [1] 281.3595
```

```
quantile(ests.np, probs=0.99, CI.level = 0.95)
```

```
## (original) estimated quantiles for each specified probability (non-censored data)
##             p=0.99
## estimate 281.3595
## Median of bootstrap estimates
##             p=0.99
## estimate 279.4572
##
## two-sided 95 % CI of each quantile
##           p=0.99
## 2.5 %  252.7795
## 97.5 % 309.5707
```

```
quantile(ests.p, probs=0.99, CI.level = 0.95)
```

```
## (original) estimated quantiles for each specified probability (non-censored data)
##             p=0.99
## estimate 281.3595
## Median of bootstrap estimates
##             p=0.99
## estimate 280.7321
##
## two-sided 95 % CI of each quantile
##           p=0.99
## 2.5 %  254.9497
## 97.5 % 311.0833
```

Exercise : TcCB Data

(Source: Millard, S.P. and Neerchal, N.K. (2001) *Environmental Statistics with S-PLUS*, CRC Press)

The guidance document *{Statistical Methods for Evaluating the Attainment of Cleanup Standards*, Volume 3: Reference-Based Standards for Soils and Solid Media (USEPA 1994b, pp. 6.22-6.25) contains 124 measures of 1,2,3,4- Tetrachlorobenzene (TcCB) concentrations (ppb) from soil samples at a Reference site and a Cleanup area (Table 1.1).

47 observations from the Reference site and 77 in the Cleanup area. These data are stored in the file `EPA.94b.tccb.df.txt`.

1. Import the dataset into R and create a dataframe named `EPA.94b.tccb.df`

In the dataset this observation is treated as censored at an assumed detection limit of 0.09 ppb (the smallest observed value).

There is one observation coded as "ND" in this data set as presented in the guidance document.

2. Select the TcCB concentrations for the Cleanup area

3. Consider a Box-Cox transformation of the selected TcCB concentrations such that the distribution of the transformed data looks similar to a Gaussian distribution (see lab3 for an example of R implementation of the tranformation)
4. Fit a Gaussian distribution over the tranformed data.
5. Found the value (on the new scale) such that is overcome 1/100 times
6. Give the corresponding value on the ppb and name this a.
7. Suppose that you take three measures from the Cleanup Area. What is the probability to obtain at least one sample that overcome the value a ?